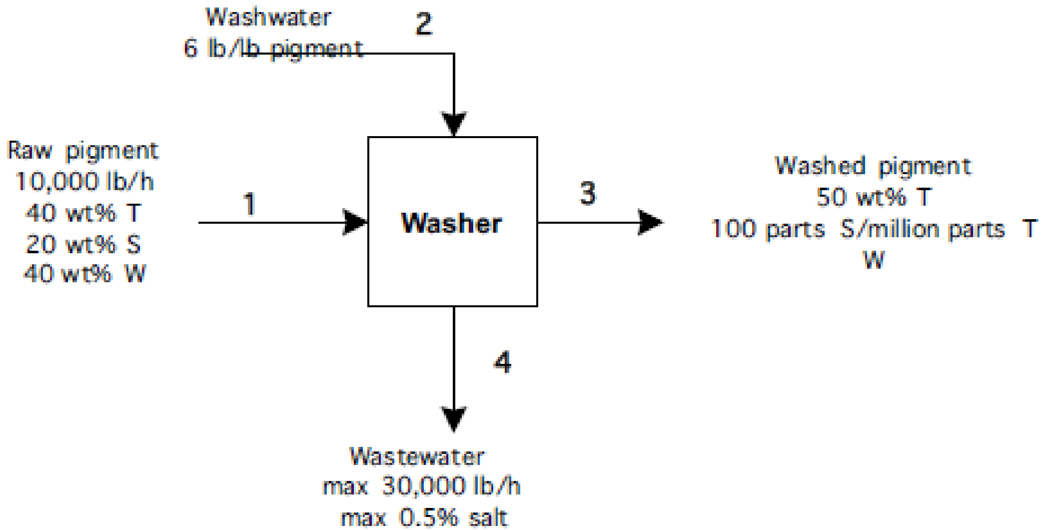
P2.66



There are 3 components: T (titanium dioxide), S (salt) and W (water). All units are in lb/h or wt%, so no conversions needed.

DOF analysis:

	Variables	Constraints
Stream	9	
System	0	
Specified flows		2 (raw pigment, wastewater)
Specified compositions		7 (2 for raw pigment, 1 for wastewater, 2 for washed pigment, 1 for ratio of washwater to raw pigment)
Specified system performance		0
Material balance		3

DOF = 9 – 12 = - 3.

The problem is overspecified, but this conclusion does depend on what we count as constraints. Several of the constraints are "soft" – that is, they are maximums or optimums, not absolute requirements.

One way to attack an overspecified problem such as this is to ignore the soft constraints (at first), complete the process flow calculations in the absence of these constraints, and then check to insure that you have not violated any of the soft constraints.

Here, we will (for now) ignore the constraints imposed on wastewater quantity and quality. We'll keep the constraint on salt concentration in the washed pigment, and on the optimum washwater feed rate. The material balance equations become:

TiO_2: $T_1 = 0.4(10{,}000) = 4000 \text{ lb/h} = T_3$

Salt: $S_1 = 0.2(10{,}000) = 2000 \text{ lb/h} = S_3 + S_4$

Water: $W_1 + W_2 = 0.4(10{,}000) + 6(10{,}000) = 64{,}000 \text{ lb/h} = W_3 + W_4$

From stream composition specifications on the washed pigment:

$$\frac{S_3}{T_3} = \frac{100}{1000000},\ S_3 = \frac{100}{1000000}(4000) = 0.4 \text{ lb/h}$$

$$\frac{T_3}{T_3 + S_3 + W_3} = 0.5 = \frac{4000}{4000 + 0.4 + W_3},\ W_3 = 3999.6 \text{ lb/h (close enough)}$$

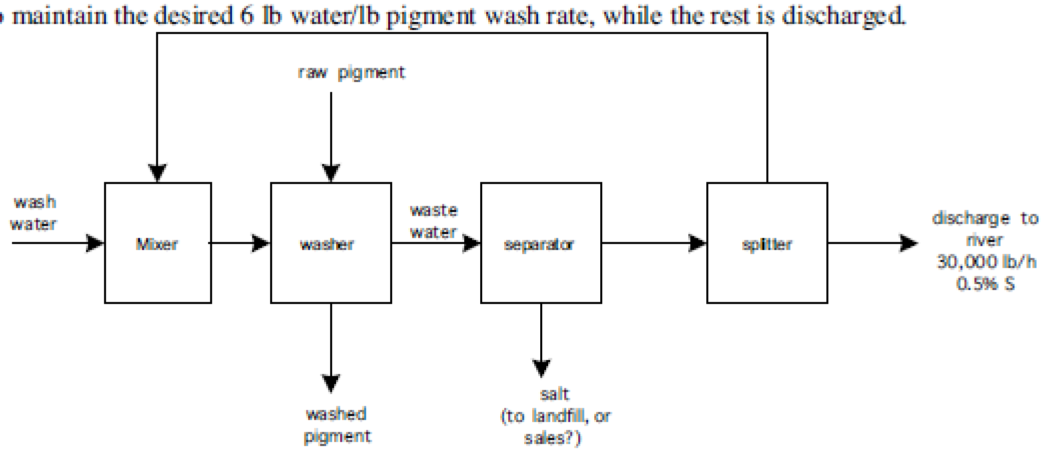
Returning to the material balances:

$S_4 = 2000 - 0.4 = 1999.6 \text{ lb/h}$

$W_4 = 64000 - 3999.6 = 60{,}000 \text{ lb/h}$

The salt concentration in the wastewater is 3.2 wt%. Neither the total wastewater discharge rate of 30,000 lb/h, or the maximum wastewater salt concentration of 0.5% is met with this design.

The only solution is to modify the design. A separation unit is required, that removes most of the salt from the wastewater and recycles part (or all) of the water. A reverse osmosis unit might work. One idea is sketched below. Excess salt is removed in the separator, to reduce the salt content of the water to 0.5 wt%. Then, part of the water is recycled and mixed with fresh water, to maintain the desired 6 lb water/lb pigment wash rate, while the rest is discharged.



P 2.69

DOF analysis:

	Variables	Explanation	Constraints	Explanation
Stream	25	3 in stream 1 3 in stream 2 5 in stream 3 3 in stream 4 2 in stream 5 1 in stream 6 1 in stream 7 1 in stream 8 2 in stream 9 2 in stream 10 2 in stream 11		
System	2	2 chemical reactions; steady-state, so no accumulation		
Specified flows			1	Cumene production rate
Specified compositions			4	95/5 P:I in stream 1 1.2:1 P:B in stream 1 10% P in stream 2 1 splitter restriction
Specified system performance			2	80% conversion of P 72% conversion of B
Material balance			20	3 for mixer 5 for reactor 5 for V/L separator 2 for splitter 3 for dist col 1 2 for dist col 2
total	27		27	

DOF = 27 – 27 = 0

The set of equations includes 25 stream variables. There are 2 reactions, each of which involve 3 compounds, so we will write 4 equations relating 6 generation and consumption variables and their stoichiometric coefficients.

Here are the equations:

Basis: $C_6 = 25$

Reaction stoichiometry: $\dfrac{C_{gen1}}{B_{cons1}} = \dfrac{1}{1},\ \dfrac{P_{cons1}}{B_{cons1}} = \dfrac{1}{1}$

$\dfrac{D_{gen2}}{C_{cons2}} = \dfrac{1}{1},\ \dfrac{P_{cons2}}{C_{cons2}} = \dfrac{1}{1}$

Stream composition: $\dfrac{P_1}{I_1} = \dfrac{95}{5}$

$\dfrac{P_1}{B_1} = \dfrac{1.2}{1}$

$\dfrac{I_2}{I_2 + P_2 + B_2} = 0.1$

Splitter: $\dfrac{P_9}{I_9} = \dfrac{P_{10}}{I_{10}}$

System performance: $P_{cons1} + P_{cons2} = 0.8P_2$

$B_{cons1} = 0.72B_2$

Material balances:

mixer
$P_2 = P_1 + P_{10}$
$I_2 = I_1 + I_{10}$
$B_2 = B_1 + B_8$

reactor
$P_3 = P_2 - P_{cons1} - P_{cons2}$
$I_3 = I_2$
$B_3 = B_2 - B_{cons1}$
$C_3 = C_{gen1} - C_{cons2}$
$D_3 = D_{gen2}$

V/L separator:
$P_9 = P_3$
$I_9 = I_3$
$B_4 = B_3$
$C_4 = C_3$
$D_4 = D_3$

splitter
$P_{10} + P_{11} = P_9$
$I_{10} + I_{11} = I_9$

dist col 1
$B_8 = B_4$
$C_5 = C_4$
$D_5 = D_4$

dist col 2
$C_6 = C_5$
$D_7 = D_5$